# Pranav Kulkarni

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## SUMMARY

Computer Science and Engineering undergraduate at B.M.S. College of Engineering with a Diploma in CSE (CGPA 9.55/10). Builds full-stack and AI-powered applications in Python, Java, React and TypeScript, including a three-factor face-and-voice biometric authentication system with a 49-endpoint FastAPI backend and CI-tested security controls. Seeking Software Engineering Internship roles.

## EDUCATION

**B.M.S. College of Engineering**, Bengaluru — Sep 2026 – Expected 2029
*Bachelor of Engineering in Computer Science and Engineering*

**Vasantrao Potdar Polytechnic**, Belagavi — Jul 2023 – Jun 2026
*Diploma in Computer Science and Engineering* — CGPA: 9.55/10 (A+)

Relevant coursework: Data Structures, Object-Oriented Programming

## TECHNICAL SKILLS

**Languages:** Python, Java, TypeScript, JavaScript, HTML, CSS
**Frameworks and Libraries:** React, Vite, Tailwind CSS, FastAPI, Pydantic, PyTorch, OpenCV, ONNX Runtime
**AI and Machine Learning:** Machine Learning, Computer Vision, Speech-to-Text, Voice Biometrics, Large Language Models (LLMs)
**Security:** Biometric Authentication, JWT, OAuth 2.0, Encryption at Rest (Fernet), Role-Based Access Control, Rate Limiting, Audit Logging
**Tools and Practices:** REST APIs, SQLite, Git, GitHub Actions (CI), Docker, Unit Testing

## PROJECTS

**BioAuth – Face and Voice Biometric Passwordless Authentication** | Project Walkthrough — Dec 2025 – Feb 2026
*React, TypeScript, Python, FastAPI, PyTorch, InsightFace, SpeechBrain, faster-whisper, SQLite, Docker*

- Built a full-stack passwordless login that requires three factors to pass: a face match (InsightFace), speaker verification (SpeechBrain ECAPA) and a spoken secret phrase transcribed with faster-whisper and fuzzy-matched.
- Designed a FastAPI backend with 49 REST endpoints, JWT sessions, three-tier role-based access, per-client rate limiting and progressive lockouts (10 min, 30 min, 2 h) against brute-force attacks.
- Protected biometric data with Fernet encryption at rest, stored secret phrases only as hashes and HMAC tokens, and added consent-based enrollment, automatic data retention cleanup and a full audit trail.
- Wrote 60+ automated backend tests run in a GitHub Actions CI pipeline; models run on CUDA GPUs with CPU fallback, and the app is containerized with Docker for cloud deployment.
- Integrated Google OAuth sign-in and built an admin console and a Gemini-powered security assistant with automatic fallback across three AI providers.

**PawCare – Stray Animal Rescue and Adoption Platform** (Team project, 4 members) — Oct 2026 – Present
*React, Node.js, Express, MongoDB, JWT, Leaflet*

- Team-built platform connecting people who report stray or injured animals with volunteers, rescuers, shelters and adopters, with role-based workflows for each user group.
- Owned testing and quality assurance: tested the app end to end and found structural errors and functional flaws, including a security gap where accounts were never verified, so fake users could act as legitimate ones.
- Reported issues to the team and contributed to the frontend design.

**Facial Emotion Recognition** — *Python, Machine Learning, Computer Vision*

- Developed a computer vision application that detects faces and classifies their emotions in real time.

**House Price Prediction and Real Estate Intelligence System** (In Progress) — *Python, Data Analysis*

- Collecting, cleaning and analyzing real estate data and building visualizations of the factors that drive house prices, as the foundation for a price prediction model.

**Exam Seating Arrangement System**

- Built a tool that handles exam scheduling and generates seating arrangements automatically using custom allocation logic.

## CERTIFICATIONS

**Artificial Intelligence** and **Cyber Security Foundation** – Infosys Springboard (Aug 2026) · **Digital Productivity with AI** – UNICEF YuWaah (Aug 2026, score 100%) · View certificates

## LANGUAGES

English (Professional Working), Hindi (Full Professional), Marathi (Native)